Classwork Set B

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Day 1 Problem 8

(a) $c = a - b$.

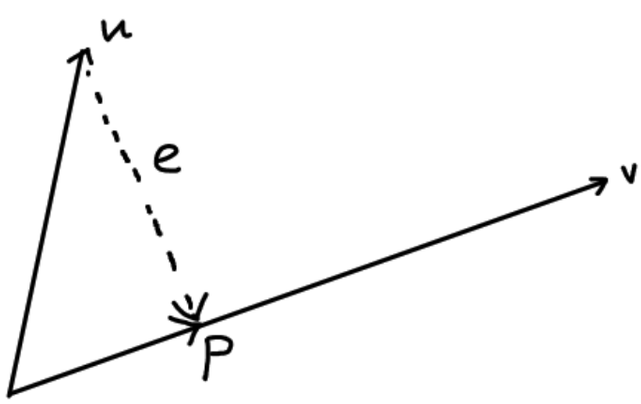
(b) Let θ be the angle between vectors a and b.

$$\begin{aligned} \|a-b\|^2 &= \|a\|^2 + \|b\|^2 - 2\|a\|\|b\|\cos\theta \\ (a-b)\cdot(a-b) &= a\cdot a + b\cdot b - 2\|a\|\|b\|\cos\theta \\ \cancel{a\cdot a} - 2a\cdot b + \cancel{b\cdot b} &= \cancel{a\cdot a} + \cancel{b\cdot b} - 2\|a\|\|b\|\cos\theta \\ \boxed{a\cdot b &= \|a\|\|b\|\cos\theta} \end{aligned}$$

Day 2 Problem 5

Let $OA, OB, OC \in \mathbb{R}^n$ be vectors with tails at the origin and heads at the poitns A, B, C, respectively. These vectors are collinear (and span a line) if and only if they do not form a linearly independent set. For example, the vector OC cannot be expressed as a linear combination of vectors OA and OB; there do not exist scalars $a, b \in \mathbb{R}$ such that $a(OA) + b(OB) = OC$.

Day 2 Problem 7



In the picture above, we have $p = \text{proj}_v(u)$, and vector e is the error between the the projection p and the actual vector u: $u + e = p \iff e = p - u$.

The projection p is a scalar multiple of vector v. Let $m \in \mathbb{R}$ be the scalar such that $p = mv$.

Notice that because this is an orthogonal projection, the projection error $e = p - u$ is orthogonal to v:

$$e \cdot v = 0 \iff (mv - u)\cdot v = 0 \iff m(v\cdot v) = u\cdot v \iff m = \frac{u\cdot v}{v\cdot v}$$

With the scalar multiple m, we can find the projection p:

$$\boxed{p = mv = \frac{u\cdot v}{v\cdot v}v = \frac{u\cdot v}{\|v\|^2}v}$$

Day 2 Problem 6

$$\text{proj}_v(u) = \frac{u\cdot v}{v\cdot v}v = \frac{(3,5)\cdot(6,2)}{(6,2)\cdot(6,2)}(6,2) = \frac{18+10}{40}(6,2) = \frac{7}{10}(6,2) = \boxed{\left(\frac{21}{5}, \frac{7}{5}\right)}$$